# Lecture 2: Basic Statistics and Review of Probability Results

## 1 Announcements

- Homework 1 has been posted. It is due on January 31st.

## 2 What is Data Analysis?

Data analysis is a process which can be split into four stages: (1) Data Collection, (2) Data Cleaning, (3) Data visualization, and (4) Statistical Data Analysis. Our focus will be on the stage (4). We begin by studying statistical techniques used to estimate mean and variance of random variables, to infer the parameter of a model from observations, and to create confidence intervals around our estimates. Before doing so, we recall some notion of convergence of random variables, as well as two fundamental results in probability (law of large numbers and central limit theorem).

### 2.1 Convergence of Random Variables

This section provides a very basic introduction to the various convergence notions for sequences of random variables. Students who would like to see the proofs of these results and additional examples are referred to `http://www.princeton.edu/~rvan/acm217/ACM217.pdf`, Section 1.5.

Let $(\Omega, \mathbb{F}, \mathbb{P})$ be a probability space. A sequence of random variables $\{X_n\}_{n\geq 1}$ *converges almost surely* to a random variable $X$ in $(\Omega, \mathbb{F}, \mathbb{P})$ if

$$\lim_{n\to\infty} X_n(\omega) = X(\omega),$$

with probability 1, i.e. $\mathbb{P}$-almost surely. We will use the notation $X_n \to X$ to denote this a.s. convergence.

A sequence of random variables $\{X_n\}_{n\geq 1}$ *converges in probability* to a random variable $X$ in $(\Omega, \mathbb{F}, \mathbb{P})$ if, for any $\delta > 0$,

$$\lim_{n\to\infty} \mathbb{P}\left(|X_n - X| > \delta\right) = 0.$$

We will use the notation $X_n \xrightarrow{\mathbb{P}} X$ to denote convergence in probability.

**Result 1:** Almost sure convergence implies convergence in probability. The converse is not true, however. To convince yourself, consider the sequence $X_n = 1$ with probability $1/n$, and zero otherwise. Suppose that $X_n$'s are independent. Then $\lim_{n\to\infty} \mathbb{P}\left(|X_n| > \delta\right) = \lim_{n\to\infty} \frac{1}{n} = 0$. This means that the sequence $\{X_n\}$ converges to zero in probability. However, because $\sum_{n\geq 1} \mathbb{P}(X_n = 1) = \infty$ and the events $\{X_n\}$ are independent, the second Borel Cantelli lemma guarantees that for any $\omega$ and $N$, there exists $m, n \geq N$ such that $X_n(\omega) = 1$ and $X_m(\omega) = 0$.

**Remark 2.1.** *The second Borel-Cantelli lemma states: If the events $E_n$ are independent and the sum of the probabilities of the $E_n$ diverges to infinity, then the probability that infinitely many of them occur is 1. Indeed, in our case, set $E_n := \{X_n = 1\}$. Then, $\sum_{n=1}^{\infty} \mathbb{P}(E_n) = \infty$, and hence the event $\{X_n = 1\}$ occurs infinitely often.*

A sequence of random variables $\{X_n\}_{n\geq 1}$ with cumulative probability distribution functions $\{F_n\}_{n\geq 1}$ *converges in distribution* to a random variable $X$ with distribution $F$ if for any bounded continuous function $g$,

$$\lim_{n\to\infty} \int g(x)dF_n(x) = \int g(x)dF(x)$$

We will use the notation $X_n \xrightarrow{d} X$, or $F_n \to F$ to denote this convergence.

**Result 2:** Convergence in probability implies convergence in distribution. The converse is not true, however. Consider the random variables $X$, $X_n$ to be i.i.d. and nonconstant random variables, in which case $X_n \xrightarrow{d} X$. Fix some $\epsilon$. Then, $\mathbb{P}(|X_n - X| \geq \epsilon)$ is positive and the same for all $n$, showing that $X_n$ does not converge to $X$ in probability.

**Theorem 2.2 (Slutsky's theorem).** *Suppose $X_n \xrightarrow{d} X$, $Y_n \xrightarrow{\mathbb{P}} c$ for some constant $c$. Then, $X_n + Y_n \xrightarrow{d} X + c$, and $X_n Y_n \xrightarrow{d} cX$. Furthermore, if $c \neq 0$, then $X_n/Y_n \xrightarrow{d} X/c$.*

# 3 LLN and Central Limit Theorem

We give two fundamental results in probability, the law of large numbers and the central limit theorem. The law of large numbers characterizes the average behavior of a large sequence of independent identically distributed random variables. The central limit theorem characterizes the fluctuation of these averages.

**Strong Law of Large numbers**: Suppose $\{X_n\}_{n\geq 1}$ is a sequence of independent and identically distributed random variables such that $\mathbb{E}[X_i] = \mu$, and $\mathbb{E}[|X_i|] < \infty$. Then

$$\frac{1}{n}\sum_{i=1}^{n} X_i \to \mu,$$

i.e., we have almost sure convergence of the sample mean $\frac{1}{n}\sum_{i=1}^{n} X_i$ to its actual mean $\mu$.

**Central Limit Theorem**: Suppose $\{X_n\}_{n\geq 1}$ is a sequence of independent and identically distributed random variables with mean $\mathbb{E}[X_i] = \mu < \infty$ and variance $Var(X_i) = \sigma^2 < \infty$. Then

$$\frac{\frac{1}{n}\sum_{i=1}^{n} X_i - \mu}{\frac{\sigma}{\sqrt{n}}}$$

converges in distribution to the standard Gaussian distribution (zero mean and unit variance).

# 4 Estimators and Estimates

The central objective of statistical inference is to make valid generalization from *samples*. The typical setup is as follows: we have a collection of items (subjects, measurements) about which we want to have information. For instance, the *population* can consist of US voters and we are interested in the question: what fraction of the population is in favor of candidate $A$? Or, the population can consist of US employees and we are interested in the question: what is the average income? In order to answer these questions, we collect items from the large population, also referred to as samples (e.g. collection by sample surveys). The fraction of the population in favor of $A$, and the average income, are the unknown *parameters*. An estimator is a function of the sample. It is used to estimate the parameter of interest.

**Example:** The estimator of the average income is the sample mean $\frac{1}{n}\sum_{i=1}^{n} X_i$, where $X_i$ is the income of the $i$th employee. The sample mean is basically the average of the data.

An important question is: what is the distribution of the statistics? Before samples are taken, they are unknown and random, so the estimator can be viewed as a random variable, and consequently admits a distribution. This is the distribution of the values of the statistics in all possible samples of a *fixed size* from the same population.

## 4.1 Consistency and Unbiasedness

An estimator $\hat{\theta} := f(X_1, X_2, \ldots, X_n)$ of a parameter $\theta$ is *unbiased* if $\mathbb{E}\left[f(X_1, \ldots, X_n)\right] = \theta$. The *bias* of an estimator is $Bias(\hat{\theta}) := \mathbb{E}\left[f(X_1, \ldots, X_n)\right] - \theta$. An estimator $f(X_1, X_2, \ldots, X_n)$ of a parameter $\theta$ is *consistent* if it converges to $\theta$ in probability. For instance, the sample mean $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ is an unbiased and consistent estimator of the population mean. Indeed, let $X_i$, $i \geq 0$, be i.i.d. random variables with mean $\mu$ and variance $\sigma^2$. Then $\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \mu$, i.e. it is unbiased. To prove it is consistent, recall that almost sure convergence implies convergence in probability. The strong law of large numbers implies what is referred to as the *weak law* of large numbers, i.e. $\bar{X} := \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{\mathbb{P}} \mu$, i.e. that the sample mean is a consistent estimator of the population mean $\mu$.

We can also ask other questions: for instance, what is the variance of $\bar{X}$? It is clear that $Var\left(\bar{X}\right) = \frac{1}{n^2}\sum_{i=1}^{n} Var(X_i) = \frac{\sigma^2}{n}$, where we have assumed that $X_i$'s are i.i.d.

Consider now the *sample variance*, defined by

$$s_X^2 := \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2,$$

where $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$. This is a consistent and unbiased estimator of $\sigma^2$. (homework problem).